

Piyusha Akash

Offensive Security Engineer

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SUMMARY

Offensive Security Engineer and low-level security researcher specializing in Windows internals, reverse engineering, exploit development, penetration testing and custom tooling. Ranked in the Top 5% globally on TryHackMe, securing 1st place in the Silver and Bronze leagues, 11th place in the Gold league, and holding a #1 weekly ranking in Sri Lanka. Author of independent vulnerability research on SetupAPI callback redirection for EDR evasion, alongside developer of offensive security frameworks including Artemis and the Native Adversary Framework in C and x86-64 Assembly. Certified in Binary Fuzzing and Reversing (CBFRPro), Penetration Testing (CAPT), and Windows Offensive APIs. Seeking an offensive security engineering, red teaming, or vulnerability research role to deploy low-level tradecraft, payload engineering, and adversary emulation and penetration testing.

TECHNICAL SKILLS

Languages: C/C++, x86-64 Assembly, Bash Scripting

Offensive Security: Reverse Engineering, Windows Internals, AV/EDR Internals, AV/EDR Evasion, Exploit Development, Offensive Tool Engineering, Malware Analysis, Red Teaming, Penetration Testing

Security Assessment: Vulnerability Assessment, OWASP Top 10, OSINT, Social Engineering, Privilege Escalation, Post-Exploitation, Enumeration, Active Directory, MITRE ATT&CK

Security Tools: Nmap, Metasploit Framework, MSFvenom, Burp Suite, Wireshark, Wazuh, Splunk, SIEM, Nessus, OpenVAS, FFUF, Netcat, Scapy, Social Engineering Toolkit (SET)

RE & Analysis Tools: IDA Pro, Ghidra, x64dbg, CFF Explorer, Microsoft Sysinternals Suite

Systems & Platforms: Windows, Kali Linux, Linux (Debian), Virtualization, Docker, Git

CERTIFICATIONS

Certified Binary Fuzzing & Reversing Professional (CBFRPro) – SecOps Group

Issued Jul 2026

Certified Associate Penetration Tester (CAPT) – Hackviser

Issued Feb 2026

Certified Windows Offensive API – Red Team Leaders

Issued 2026

Introduction to Linux LFS101 – The Linux Foundation

Issued 2026

COMPETITIVE SECURITY & TRAINING

TryHackMe – Practical Cybersecurity & Competitive Hacking

Dec. 2024 – Present

Hands-on Penetration Testing and Threat Hunting Platform

- Ranked in the **Top 5%** globally; secured **1st place** in Silver and Bronze leagues, **11th place** in Gold league, and **#1 weekly ranking** in Sri Lanka
- Completed 142+ hands-on labs covering penetration testing, memory forensics, OS hardening, and vulnerability chaining
- Skills: Linux exploitation, memory forensics, network scanning, Nmap, Burp Suite, FFUF, Netcat

Practical Low Level Exploitation & Reverse Engineering

Apr. 2026 – Present

PWN College

- Advanced low-level exploitation and reverse engineering training in sandboxed environments
- Built custom shellcode and ROP chains; studied memory corruption, privilege escalation, and OS architecture

Network Technician – NVQ Level 4

Jan. 2025 – Present

Technical College, Gampaha

Gampaha, Sri Lanka

- Information and Communication Technologies; networking fundamentals and applied network security

PROJECTS

- Artemis** | *C, x86-64 Assembly, Windows NT* 2026 – Present
- Low-level Windows NT offensive framework featuring a syscall resolver supporting 14,000+ Windows builds
 - Implemented inline and indirect syscall stubs engineered for EDR telemetry evasion
 - Integrated XOR, Rolling XOR, and RC4 payload encryption schemes
 - github.com/0x3xp/Artemis
- Native Adversary Framework** | *C, x86-64 Assembly* 2026 – Present
- Windows process injection framework implementing process hollowing and thread hijacking techniques
 - PEB walking and API hashing implemented entirely in x64 Assembly for stealth and portability
 - Custom PE parsers for IAT/EAT analysis enabling dynamic import resolution without Windows API
 - github.com/0x3xp/Native-Adversary-Framework
- SENTINEX** | *C, Windows x64* 2026 – Present
- Stealth-focused, modular Windows reconnaissance framework designed for red team operations and adversary simulation
 - Features a low-noise operator shell with modular recon commands, AV/EDR exposure awareness, and LOLBins discovery
 - Covers Windows system, network, and security enumeration with minimal operational footprint
 - github.com/0x3xp/SENTINEX

RESEARCH & VULNERABILITY DISCLOSURE

- Control Flow Redirection via SetupAPI Callback** | *Independent Security Research* 2026
- Documented methodology to subvert `setupapi.dll` callback routines using executable memory regions, bypassing endpoint detection telemetry that monitors conventional execution APIs
 - Published a C proof-of-concept alongside defensive mitigation strategies
 - [github.com/0x3xp/\[...\]/SetupAPI-Callback-Redirection](https://github.com/0x3xp/[...]/SetupAPI-Callback-Redirection)
- Microsoft Defender Security Bypass** | *Responsible Disclosure – MSRC* 2026
- Identified a novel security bypass vulnerability in Microsoft Defender; currently under active review by the Microsoft Security Response Center (MSRC)
 - Full technical details remain confidential pending vendor remediation and coordinated public disclosure

REFERENCES

The Principal

Technical College, Gampaha

Institutional referee confirming course enrollment, academic compliance, and performance standards

Phone: (+94) 332237579

Malinda Fernando

Lecturer, Technical College, Gampaha

Academic referee for practical technical skills, project development, and laboratory performance

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